

a. Metabolites included in the Metabolic Risk Score (MetRS)

Metabolite	Estimate (95% CI)	p-value	Variance explained	HMDB ID	Subclass	Direct parent
Metabolites included in MetRS						
3-methyladipate/pimelate*	-4.48 (-8.11, -0.86)	1.56E-02	10.54	HMDB00555	Fatty acids and conjugates	Medium-chain fatty acids
13-HODE*	-3.66 (-7.03, -0.29)	3.36E-02	6.88	HMDB04667	Lineolic acids and derivatives	Lineolic acids and derivatives
glucuronate*	-4.27 (-7.91, -0.63)	2.17E-02	5.04	HMDB00127	Carbohydrates and conjugates	Glucuronic acid derivatives
2-aminoisobutyric acid*	6.26 (3.01, 9.50)	2.03E-04	6.28	HMDB01906	Amino acids, peptides, and analogues	Alpha amino acids
Kynurenine	9.23 (5.78, 12.67)	4.36E-07	4.04	HMDB00684	Carbonyl compounds	Alkyl-phenylketones
Homoarginine	5.86 (2.52, 9.21)	6.98E-04	2.90	HMDB00670	Amino acids, peptides, and analogues	L-alpha-amino acids
Hydroxyproline	-4.29 (-7.65, -0.94)	1.26E-02	2.84	HMDB00725	Amino acids, peptides, and analogues	Proline and derivatives
Malonate	4.96 (1.65, 8.27)	3.54E-03	2.47	HMDB00691	Dicarboxylic acids and derivatives	Dicarboxylic acids and derivatives
1-methylguanosine	-3.82 (-7.05, -0.60)	2.03E-02	2.04	HMDB01563	Purine nucleosides	Purine nucleosides
4-Hydroxyhippurate	-3.24 (-6.78, 0.30)	7.26E-02	1.26	HMDB13678	Benzoic acids and derivatives	Hippuric acids
N-Acetylserine	3.80 (-0.18, 7.79)	6.13E-02	1.20	HMDB02931	Amino acids, peptides, and analogues	N-acyl-L-alpha-amino acids
N-Carbamoyl-beta-alanine	-2.74 (-6.29, 0.82)	1.30E-01	0.84	HMDB00026	Ureas	Ureas
C34:2 PC	-0.02 (-0.06, 0.01)	1.68E-01	0.23	HMDB07973	Glycerophosphocholines	Glycerophospholipids
Metabolite not included in MetRS						
DMGV*	-7.26 (3.4,11.1)	2.33E-04	0.082	HMDB24021	Short-chain keto acids and derivatives	Short-chain keto acids and derivatives

